

# Advanced Machine Learning

DATA 642

Lecture 1 — Basic Vector Calculus

\*Figures and notes are from the book Mathematics for Machine Learning by A. Aldo Faisal, Cheng Soon Ong, and Marc Peter Deisenroth

# Basics

## Gradient

- The generalization of the derivative to functions of several variables is the gradient. We find the gradient of the function  $J$  with respect to  $\boldsymbol{\theta}$  by varying one variable at a time and keeping the others constant. The gradient is then the collection of these partial derivatives.
- For a function  $J(\boldsymbol{\theta}) : \mathbb{R}^l \rightarrow \mathbb{R}$  of  $l$  variables we define the **partial derivatives** as

$$\frac{\partial J}{\partial \theta_i}, i = 1, \dots, l \quad (1)$$

If we collect all partial derivatives in a row vector we create the gradient of  $J$ . We denote the gradient of  $J$  as

$$\nabla J = \left[ \frac{\partial J}{\partial \theta_1} \frac{\partial J}{\partial \theta_2} \cdots \frac{\partial J}{\partial \theta_l} \right] \in \mathbb{R}^{1 \times l} \quad (2)$$

## Chain rule

- Let us consider a function  $J : \mathbb{R}^2 \rightarrow \mathbb{R}$  of two variables  $\theta_1, \theta_2$  where  $\theta_1(t)$  and  $\theta_2(t)$  are themselves functions of  $t$  (depend on  $t$ ). To compute the gradient of  $J$  with respect to  $t$ , we apply the chain rule for multivariate functions as

$$\frac{dJ}{dt} = \left[ \frac{\partial J}{\partial \theta_1} \frac{\partial J}{\partial \theta_2} \right] \left[ \frac{\partial \theta_1(t)}{\partial t} \frac{\partial \theta_2(t)}{\partial t} \right]^\top = \frac{\partial J}{\partial \theta_1} \frac{\partial \theta_1(t)}{\partial t} + \frac{\partial J}{\partial \theta_2} \frac{\partial \theta_2(t)}{\partial t} \quad (3)$$

## Hessian (Second order)

- Let us consider a function  $J : \mathbb{R}^2 \rightarrow \mathbb{R}$  of two variables  $\theta_1, \theta_2$ . If  $J(\theta_1, \theta_2)$  is twice differentiable the **Hessian** matrix is defined as

$$\begin{bmatrix} \frac{\partial^2 J}{\partial \theta_1^2} & \frac{\partial^2 J}{\partial \theta_1 \partial \theta_2} \\ \frac{\partial^2 J}{\partial \theta_2 \partial \theta_1} & \frac{\partial^2 J}{\partial \theta_2^2} \end{bmatrix} \quad (4)$$

Sometimes we also denote the Hessian by  $\nabla^2 J$ . The Hessian is a symmetric matrix. This is an important matrix in optimisation since it measures the **curvature** of the function  $J$  around a point  $(\theta_1, \theta_2)$ .